MILITARY SPECIFICATION
MANUALS, TECHNICAL: TESTING AIRCRAFT POWER PACKAGE
(Quick Engine Change Configuration)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

I. SCOPE

1.1 This specification covers the detail requirements for the preparation of Technical Manuals covering testing instructions for aircraft power package in quick engine change configuration with the power package removed from the aircraft. This specification shall apply when test facilities have been provisioned and the using command maintenance plans require test and trim of removed power packages.

2. APPLICABLE DOCUMENTS.

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-M-38784	Manuals, Technical: General Style and Format Requirements
MIL-M-38790	Printing Production of Technical Manuals, General Requirements For

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS.

3.1 General. The general manner of preparation shall be in accordance with MIL-M-38784 and MIL-P-38790. The manual shall be prepared in the form (manuscript, reproducible copy, negatives) specified in the contract.

3.1.1 Text. The requirements of MIL-M-38784 are applicable. The manual shall contain all essential information required to install power package in the test stand, conduct tests, adjust engine trim, analyze any malfunctions evidenced during test and remove power package from test stand. Ground safety precautions shall be stressed as necessary throughout the test.

3.1.1.1 The instructions shall include a functional test of all power package components remotely controlled by or having indicators in the cockpit.

3.1.2 Illustrations. Illustrations and diagrams shall be provided in accordance with MIL-M-38784.

3.1.2.1 Diagrams of systems, such as fuel, oil, and hydraulic, shall be used to the extent necessary to enable the user to obtain a clear understanding of the subject matter.

3.2 Arrangement. Manuals shall be arranged in the following manner. If a section(s) is not applicable to a particular manual, because of the nature of the equipment, the section number and title will be inserted, followed by "Not Applicable".

Front Matter

Section I General Information

TMSS

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Section II Special Tools

Section III Testing Facility

Section IV Pre-Operative Procedures

Section V Operating Procedures

Section VI Trouble Analysis

Section VII Post-Operative Procedures

3.3. Front Matter. Front matter shall be prepared in accordance with MIL-M-38784 and consist of: title page and list of effective pages; table of contents; list of illustrations; list of tables, (if applicable); and the introduction. The introduction shall provide a brief explanation of the manuals purpose. It shall identify the engine(s) by Air Force type, model, and series number and the aircraft on which the power package is installed by model and series number. Identification of QEC power packages in regard to installed position differences shall also be included.

3.4 Section I - General Information. This section shall contain a general introduction to the power package testing with a brief description of the test stands, engine system components to be tested, and special testing configuration.

3.5 Section II - Special Tools and Test Equipment.

3.5.1 General Requirements. This section shall list and illustrate, as necessary, the special tools and test equipment required for the work described. Items shall be selected from the list of Government approved tools and test equipment, if available, otherwise the contractor's recommended tools and test equipment shall be shown. However, these recommendations shall be changed promptly, if necessary, to conform to the official Government List as soon as such information is available. Acceptable alternate tools and test equipment shall also be listed. Standard tools, such as screwdrivers, pliers, etc., shall not be listed; however, standard items of test equipment, such as voltmeters, tube testers, etc., shall be listed, but not illustrated. Introductory material to the section shall include a statement to the effect that "equivalent" items of like function, precision, effectiveness and tolerence may be used if the recommended tool/test equipment is not available.

3.5.2 Special tools and test equipment list. Special tools and test equipment shall be listed in the following format. Special tools shall be grouped and listed first, with the list of test equipment following.

SPECIAL TOOLS AND TEST EQUIPMENT LIST

Tool/Equipment Number	Figure No.	Nomenclature	Use and Application

a. Tool/equipment number. Tools shall be listed in sequence by part (tool) number, which shall agree with the drawing number. Test equipment shall be listed in alpha-numeric sequence by type designation, using the AN type designation, if assigned, or the commercial or manufacturer's designation if an AN type designation has not been assigned. If an alternate item can be used, it shall be listed immediately following the item for which it is an alternate and shall be so identified as an alternate.

b. Figure number. This column shall contain the number of the applicable illustration in which the tool/equipment is shown.

c. Nomenclature. The nomenclature of the tool/equipment shall be the standard nomenclature and agree with the drawing title.

d. Use and application. The purpose of each item shall be stated. If use is described in the manual, reference to the applicable paragraph will suffice. If the manual covers more than one model and a special tool/equipment is applicable to particular model(s), the application shall be shown in this column.

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3.5.3 Illustrations of special tools and test equipment. Special tools and test equipment shall be illustrated in this section unless they are adequately illustrated in use in other sections of the manual. Standard items need not be illustrated as such, nor shall self-evident or generally known uses be shown. Uncommon or unusual uses and connections for test purposes shall be illustrated in appropriate sections to avoid misunderstanding as to proper procedures. Unusual operations in checking fits, clearances, backlash, etc., shall be illustrated.

3.5.4 Tool and test equipment maintenance. Required maintenance instructions for peculiar tools and test equipment, not covered in other manuals, shall be included in this section. Instructions shall also be included for fabricating interconnecting cable assemblies, etc., required but not supplied with the equipment.

3.6 Section III - Testing Facility. This section shall include a brief description of the entire test facility and functional descriptions of the controls and indicators used and monitored during power package testing. Illustrations shall be provided to portray control panels and indicators peculiar to testing of specific power package. Calibration requirements for peculiar QEC Test Instrumentation shall include the frequency of calibration and required accuracy of instruments only. Refer to pertinent instrument Technical Orders for calibration instructions.

3.7 Section IV - Pre-Operative Procedures. This section shall contain detailed step-by-step instructions for all preliminary operations required prior to performing a power package test run. Instructions shall include:

(1) Initial Preparation of test area.

(2) Engine transfer from transportation dolly to run-up dolly.

(3) Installation and positioning of equipment and special tools required for test.

(4) Engine inspection.

(5) Installation of run-up dolly in test facility.

(6) Servicing instructions.

(7) Hook-up of all control and instrumentation connection (at engine and test facility) and preliminary leak check.

(8) Final preparation of test area.

(9) Positioning of power switches, circuit breakers, etc.

(10) Engine Anti-icing valves check (check both engine anit-icing valves and nose cowl anti-icing valves).

(11) Ignition system check.

(12) CDS oil system priming.

(13) Oil sampling (if applicable).

3.8 Section V - Operating Procedures. This section shall consist of step-by-step operating procedures, charts, graph, electrical schematics, and flow diagrams necessary to perform the power package test and shall include, but is not limited to, the following:

(1) Pre-trim preparation instructions.

(2) Test reference table.

(3) Temperature limit charts for engine starting, acceleration, and stablized conditions.

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(4) Trim time limitation figure.

(5) Trim Target Charts shall be prepared and properly annotated to indicate correction factors when using closed type cells, inlet screens, or sound suppressors.

(6) Chart for determining depression factors.

(7) Engine starting, shut-down, and emergency shut-down procedures.

(8) Engine fuel flow and engine oil system pressure check.

(9) Engine RPM indicating system check.

(10) Engine breather check.

(11) Alternator functional check - (Use exciting *AlA Load Bank*) *Engine* vibration will be taken during loaded condition. A token load may be required during run to prevent damage to the system.

(12) EGT Harness Check - Spread check and average will be made with existing equipment.

(13) EPR system check (if applicable).

(14) Bleed value functional check (use EPR limits).

(15) Fuel pump output pressure.

(16) PT7 Pressure Measurement

(17) *Anti-ice functional* check.

(18) Engine oil temperature measurement.

(19) Oil consumption check, curves, etc.

(20) Hydraulic pump functional check (using system comparible to M37).

(21) Generator check-out.

(22) Trim procedures in step-by-step form.

(a) Trim checks shall include simulated wet trim for temperature presently authorized.

(b) N1 and N2 rpm (engine conditioning and overspeed check).

(c) Field cleaning (carbo-blast).

(d) Vibration analysis check.

(e) EGT spread check.

3.9 Section VI - Trouble Analysis. This section shall include trouble shooting tables and procedures. Electrical schematics and flow diagrams shall be provided as required, as an aid to trouble shooting and engine system checkouts. Trouble shooting instructions shall be listed and arranged under the following headings:

Trouble Probable Cause Isolating Procedure Remedy

Remedies shall be brief and concise. The table shall also contain references to test points shown on the diagrams so that the table and diagrams combined will provide a systematic procedure for isolating the source of malfunction.

3.10 Section VII - Post -Operative procedures. This section shall contain step-by-step instructions for all operations required to restore the test facility to normal configuration after power package testing. Instructions shall include:

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(1) Removal of special equipment from power package and test facility.

(2) Removal of power package from test stand.

(3) References to applicable Technical Order for engine preservation and depreservation.

(4) Oil Sampling (if applicable).

3.11 Section VIII - Difference Data. This section shall contain Difference Data required to provide information on additional models of test facilities which constitute minor changes from the original design. Separate Difference Data sheets shall be prepared for each additional model.

3.11.1 Difference Data sheets shall specify, in the briefest practical form, how operation and test procedures for that specific model differ from procedures for the representative model contained in the preceding sections.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Quality assurance provisions shall be in accordance with requirements of MIL-M-38784 and MIL-P-38790.

5. PREPARATION FOR DELIVERY.

5.1 Packaging, packing, and markings for shipment shall be in accordance with the requirements of MIL-M-38784 and MIL-P-38790.

6. NOTES.

6.1 Intended Use. Technical manuals prepared in accordance with the requirements of this specification are intended for use by maintenance personnel for testing and trim of aircraft power packages in quick engine change configuration, with the power package removed from the aircraft.

CUSTODIAN:

Air Force - 16

REVIEWERS:

Air Force 01, 10, 11, 70, 71, 80, 82

PREPARING ACTIVITY:

Air Force - 16

PROJECT NO. TMSS-F405

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-M-83498(USAF) Manuals Technical:Test A/c Power Package

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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